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Geometric Design and Safety Aspects of Roundabouts

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The findings of a 3-year study conducted at Pennsylvania State University for FHWA on the safety and operational performance and geometric characteristics of roundabouts are presented. The study focused on geometric characteristics that influence the safety and operational performance of roundabouts. Accident data (including driver testimonials) were reviewed along with videotapes developed at several roundabouts in Maryland, Florida, and Nevada. Several conclusions were drawn, including the need to improve geometric design on approaches to rural roundabouts to reduce loss-of-control accidents; the need for adequate right-of-way to properly deflect vehicles around the center island; and the need for guidance regarding operating-volume-to-capacity ratios. The findings from the study will assist planners, designers, and engineers in avoiding unnecessary safety hazards and operational failures.

Roundabouts are being implemented throughout the United States in a variety of situations. Many states are considering roundabouts as a viable alternative to two-way stop-controlled intersections, all-way stop-controlled intersections, and, in some cases, signals and complex freeway interchanges. Because of the increasing popularity of roundabouts in this country, the Highway Capacity and Quality of Service Committee included a procedure for estimating approach capacity at single-lane roundabouts based on international experience and limited U.S. data in the newly released *Highway Capacity Manual* 2000. In addition, FHWA has released a design guide for roundabouts in the United States, *Roundabouts: An Informational Guide* (*1*), that was developed by Kittelson & Associates along with a panel of international experts.

With the heightened awareness of the viability of roundabouts as alternatives to traditional intersection design in the United States, designers and engineers need to be educated as to the proper techniques and possible errors that can be made when designing roundabouts. This paper was developed from information gathered during a 3-year study conducted at The Pennsylvania State University for FHWA on the operations, safety, and design of roundabout performance. The findings from this study will assist planners, designers, and engineers in avoiding unnecessary safety hazards and operational failures.

BACKGROUND

The study conducted at The Pennsylvania State University investigated the safety and operational performance, as well as design characteristics, of several roundabouts located in Maryland, Florida, and Nevada. Nine single-lane roundabouts located in Maryland and Florida were included in this study, along with two multilane roundabouts located in Nevada. The roundabouts chosen for this study adhere to the basic definition of a roundabout in that

- Entering traffic must yield to traffic in the circulating stream before entering the roundabout,
- Geometric elements control the entry speed of vehicles versus signs or pavement markings, and
- Vehicles are channelized on the approach before entering the roundabout (*1*).

Information was gathered for the study by reviewing hard-copy accident data (which included testimonies of drivers involved in accidents) and site plans and by conducting several site visits. During the site visits, many hours of videotape were recorded from several directions, including approaches to the roundabouts, roundabout entries, and overhead, which allowed for monitoring of circulating stream driver performance. Each site was videotaped for 4 h during the peak periods. In addition, several hours were spent at each site for pure observation to better understand the operational and safety performance of the roundabouts. Specific information on the sites included in this study is given in Table 1.

PRELIMINARY SAFETY PERFORMANCE FINDINGS

To better understand the safety performance of the roundabouts included in this study, a preliminary comparison of the before and after accident rates and frequencies was performed. Study sites that were chosen for this preliminary safety study were required to have been operational for 2 years or more and also to have accident data available for the before period. For this study, the period when the intersections were operating as roundabouts is referred to as the "after" period, and the period when the intersections were operating as minor stop-controlled intersections is referred to as the "before" period. Accident data were gathered for only eight single-lane roundabouts; no multilane roundabouts were included in accident analysis study. Accident rates and injury rates were calculated for the study sites based on the procedure presented in the *Highway Safety Improvement Program* guide and are presented in Table 2 (*2*). Given the short performance period and rural location of many of the roundabouts, it was assumed that traffic volumes did not increase at these sites after roundabouts were constructed.

From the preliminary safety study, the safety performance of the retrofitted roundabouts appears to have improved after the installation of the roundabouts. As of January 1998, the eight roundabouts included in the study were performing better in terms of reduced

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TABLE 1 Safety Study Site Information

Location	Control in Before Period	Average Daily Traffic (All Approaches)	Peak Hour Volume (All Approaches)
Palm Beach County, FL	TWSC	7,600 vpd	510 vph
Lisbon, MD	TWSC	8,500 vpd	856 vph
Tallahassee, FL	T-Intersection[a]	17,825 vpd	1,085 vph
Fort Walton Beach, FL	T-Intersection[a]	12,000 vpd	1,245 vph
Lothian, MD	TWSC	15,000 vpd	1,345 vph
Washington County, MD	TWSC	7,000 vpd	800 vph
Cecil County, MD	TWSC	6,000 vpd	800 vph
Carroll County, MD	TWSC	12,500 vpd	1,300 vph

NOTE: TWSC = two-way stop controlled; vpd = vehicles per day; vph = vehicles per hour.
[a] T-intersections included in this study had stop control on the minor approach.

accident frequency, accident rates, and injury accident rates than they did in their before periods. The findings from this safety study are consistent with the majority of findings from international safety studies (*3–5*).

The reduction in accidents at roundabouts compared with minor stop-controlled intersections can be largely attributed to the reduction in conflict points between the two geometric configurations. A conflict point is defined as a point or place in the intersection proper in which a vehicle can cross another vehicle's travel path. Figure 1 depicts the change in the number of conflict points from 32 points (at a four-leg intersection) to only 8 points when a single-lane roundabout is used as an alternative geometric configuration (*1*). Another advantage of roundabouts, compared with traditional at-grade intersections, is the use of a center island. The center island of a roundabout greatly reduces the occurrence of severe accidents such as right angle, left-turn head-on, and head-on accidents, thereby reducing the number of injury accidents. Figure 1 supports the findings drawn from the accident reports at these eight sites that the single-lane roundabouts operate better in terms of safety than they did as minor stop-controlled intersections.

EFFECT OF GEOMETRIC DESIGN ON SAFETY PERFORMANCE

The safety performance of the study sites was positive in that fewer accidents occurred after the installation of roundabouts. To identify geometric elements that may influence the safety performance of the study sites, hard-copy accident reports were reviewed. Figure 2 was developed from information included in the accident reports. Figure 2 includes a breakdown, by accident type, of the 33 accidents reported in the after period when the intersections were operated as roundabouts. As shown in Figure 2, 27.3 percent of the accidents reported were sideswipe accidents that occurred when entering drivers failed to yield to circulating drivers. Of the sideswipe accidents reported, two of every three occurred because of driver traffic violation. This type of accident may reduce as driver experience with roundabout operation increases. Further education of the public before installation of a roundabout may also help to reduce this type of accident.

Of greater importance, 45.47 percent of the total accidents reported at the study sites were classified as "loss-of-control" accidents. Of the loss-of-control accidents reported, three of every five were the result

TABLE 2 Safety Performance Data

Study Site	Performance Period	Accident Frequency Per Year (Before/After)	Accident Rate (Before/After)	Injury Accident Rate (Before/After)
Palm Beach County, FL	2 years	1.5/1.5	0.54/0.54	0.5/0.0
Lisbon, MD	2 years	7.5/2.5	2.42/0.81	1.5/0.5
Tallahassee, FL	2 years	4.5/1.5	0.69/0.23	0.0/0.0
Fort Walton Beach, FL	2 years	8.0/2.0	1.83/0.45	2.0/0.0
Lothian, MD	2 years	13.0/4.0	2.37/0.73	4.5/1.5
Washington County, MD	2 years	4.5/0.0	1.76/0.0	1.0/0.0
Cecil County, MD	2 years	3.0/0.0	1.37/0.0	1.0/0.0
Carroll County, MD	2 years	5.3/0.0	1.81/0.24	2.25/0.75

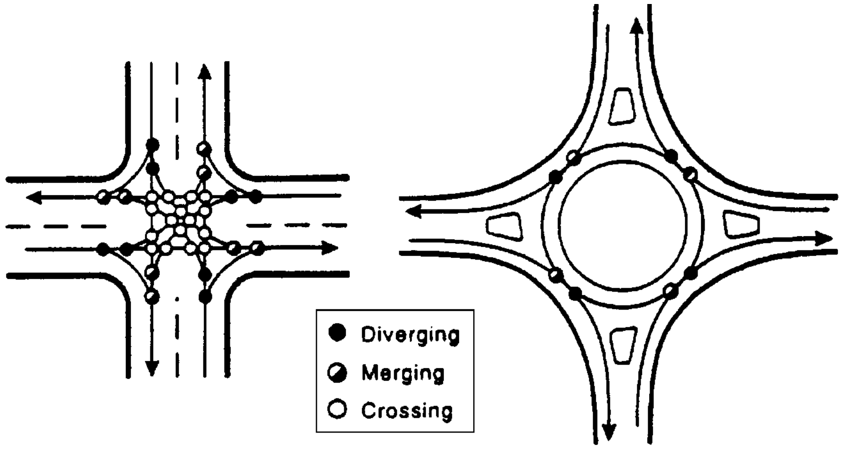

FIGURE 1 Conflict points at four-leg intersection and single-lane roundabout (*1*).

of entering drivers approaching the roundabout at excessive speeds, as stated in the accident reports. As previously mentioned, entry design is key to slowing drivers as they approach roundabouts to allow them to safely reduce their speed before entering the roundabout. Also, given that 14 of the 15 loss-of-control accidents occurred at high-speed rural road locations, the speed reduction associated with the approach curve does not appear to be adequate. It also appears that the drivers' expectations are perhaps being violated at the rural roundabouts. Straight approaches and high speeds do not convey to some of the approaching drivers that a change in the roadway geometry lies ahead, despite the use of warning signs on the approaches. In this environment, additional features may be necessary to increase driver awareness of the changing roadway characteristics.

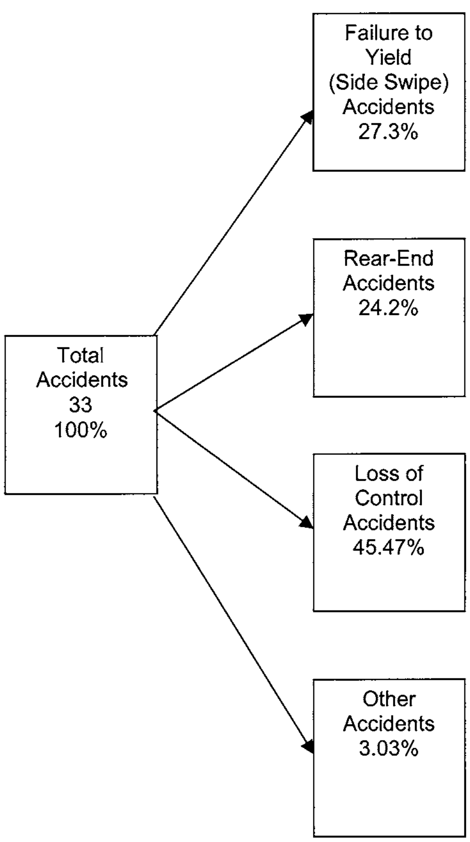

FIGURE 2 Accident types reported at single-lane roundabouts.

The Maryland roundabout design guide suggests that rumble strips be applied to high-speed approaches to roundabouts, such as rural roundabouts (*6*). Another technique suggested for approaches to high-speed rural roundabouts is the use of successive reverse curves, as shown in Figure 3. Successive reverse curves are intended to slow vehicle speeds further upstream of the approach curve so that vehicles enter the roundabout at an appropriate speed. This technique may help to reduce the failure-to-yield and loss-of-control accidents attributable to excessive speeds before the yield point of the study roundabouts. A recently completed study found that at a high-speed rural roundabout where this technique was applied, the single vehicle accident rate reduced from 0.73 accident per year without the treatment to 0.41 accident per year in the after period (*7*). Considering the number of loss-of-control accidents that occurred at rural roundabouts included in this study, the use of successive reverse curves may prove beneficial in reducing this type of accident.

Several observations can be made from the preliminary safety study of single-lane roundabouts. Roundabouts appear to perform better in terms of reduced accident frequency, accident rate, and injury rate compared with the before period performance as minor stop-controlled intersections. In addition, the in-depth review of the accident reports gathered for this study demonstrates how critical the entry speed of vehicles is in safety performance. This point was also stressed in the August 2000 publication by FHWA, *Roundabouts: An Informational Guide* (*1*). Achieving speed consistency between conflicting traffic streams to promote the design of safer roundabouts is an underlying principle in the FHWA guide. The guide recommends that the difference in design speeds be kept to within 20 km/h (12 mph) between the entry path radius and the conflicting circulating path radius. This strategy helps to increase reaction time, which may in turn reduce crash rates and crash severity (*1*).

EFFECT OF SITE SELECTION AND DESIGN VOLUMES ON OPERATIONAL PERFORMANCE

In addition to the eight single-lane roundabouts included in the safety study, several additional roundabout sites were visited as part of this overall effort. After visiting all of the potential sites and

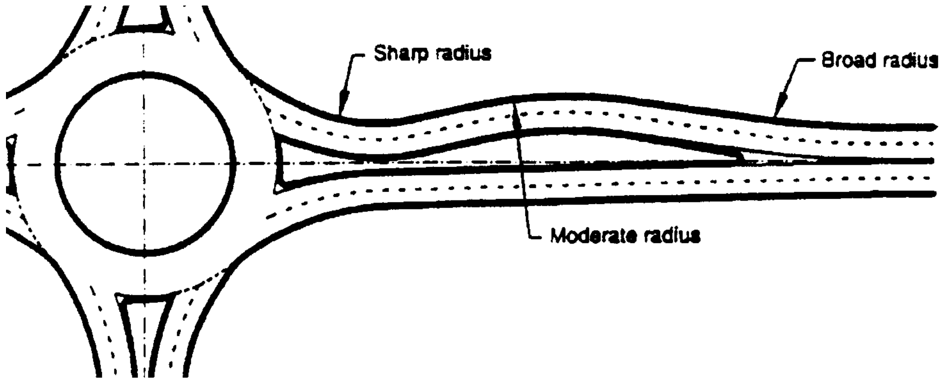

FIGURE 3 Successive curves on high-speed rural roundabout approaches (*1*).

reviewing site plans and accident data, three roundabouts were not included in the safety study. These sites were not chosen for the study because the two multilane roundabouts located in Summerlin, Nevada, lacked sufficient "before" accident data and the roundabout located in Bradenton Beach, Florida, failed to meet the basic criteria of a roundabout. Although not sufficient to be included in the safety study, the information and observations gathered at these sites did prove to be interesting and valuable. This information is presented in the form of case studies. The first case study addresses the need for adequate right-of-way to properly design a roundabout and the effect of unbalanced flows on roundabout operations. The second case study addresses the topic of design hourly volume and the problems associated with roundabouts operating at extremely low volume to capacity (v/c) ratios.

Bradenton Beach, Florida, Case Study

Just as safety performance can be hindered by inadequate geometric design, operational performance can be hindered by improper site selection. Here, site selection refers to the specific location of roundabouts in relation to intersection characteristics. A few examples of compromised geometric and operational features are addressed in this section:

- Lack of proper sight distance,
- Lack of available right-of-way for proper deflection at urban roundabouts, and
- Lack of balanced traffic flows.

Figure 4 contains a photo of a roundabout located at the intersection of Bridge Street and SR 789 in Bradenton Beach, Florida. Table 3 contains peak hour volume measurements for all approaches to the roundabout. A large building is located on the southeast corner of this intersection. Using as-built construction plans, sight distance triangles were developed for the westbound approach. The available external sight distance for this approach was determined to be 26 m from the as-built construction plans. The recommended sight distance for this approach to the upstream entering stream was determined to be 54.2 m, using the FHWA criteria for a 6.5-s critical gap and an approach speed of 30 km/h (*1*). Based on these measurements and calculations, this approach lacks an external sight distance of 28.2 m on the westbound approach. This required distance is shown in Figure 5.

The limited sight distance forces drivers to slow considerably before reaching the yield bar because they cannot see around the building to the upstream approach to scan for potential conflicts. This behavior goes against the basic operational characteristic of roundabouts: free flow of vehicles through the intersection given that an acceptable gap is available. In addition, the location of the building on the southeast corner compromises the available right-of-way, thus affecting the deflection of the northbound and southbound approaches. The result is that vehicles do not slow to enter the roundabout on the northbound and southbound approaches, and they essentially maintain their straight travel paths by driving over the truck apron. In fact, the truck apron began to fail structurally only 1 year after the roundabout was opened. From Figure 4, the truck apron can be seen as the raised area around the center island provided, in some cases, to allow larger vehicles to traverse the roundabout easily.

In addition, the flow of vehicles is very unbalanced at this site, as is shown in Table 3. The northbound and southbound approaches carry approximately 97.3 percent of the total flow through the roundabout. This imbalance of traffic flows results in very few acceptable gaps available to the westbound approach, making it increasingly difficult for drivers on this approach to enter the roundabout comfortably.

The combination of (*a*) poor sight distance on the westbound approach, (*b*) lack of deflection on the northbound and southbound approaches, and (*c*) unbalanced flows produces a very expensive minor stop-controlled intersection. This case study demonstrates the

FIGURE 4 Bradenton Beach, Florida, roundabout.

TABLE 3 Bradenton Beach, Florida, Peak Hour Volumes

Approach	Peak Hour Approach Volume
NB SR 789	194 vph
SB SR 789	445 vph
WB Bridge Street	17 vph

need for designers, planners, and engineers to understand the importance of basic roundabout requirements. There must be sufficient right-of-way to properly construct a roundabout to allow for deflection of vehicles around the center island (not requiring the use of the truck apron) and adequate sight distance to allow free flow of vehicles into the roundabout, provided an adequate gap in the circulating stream is available.

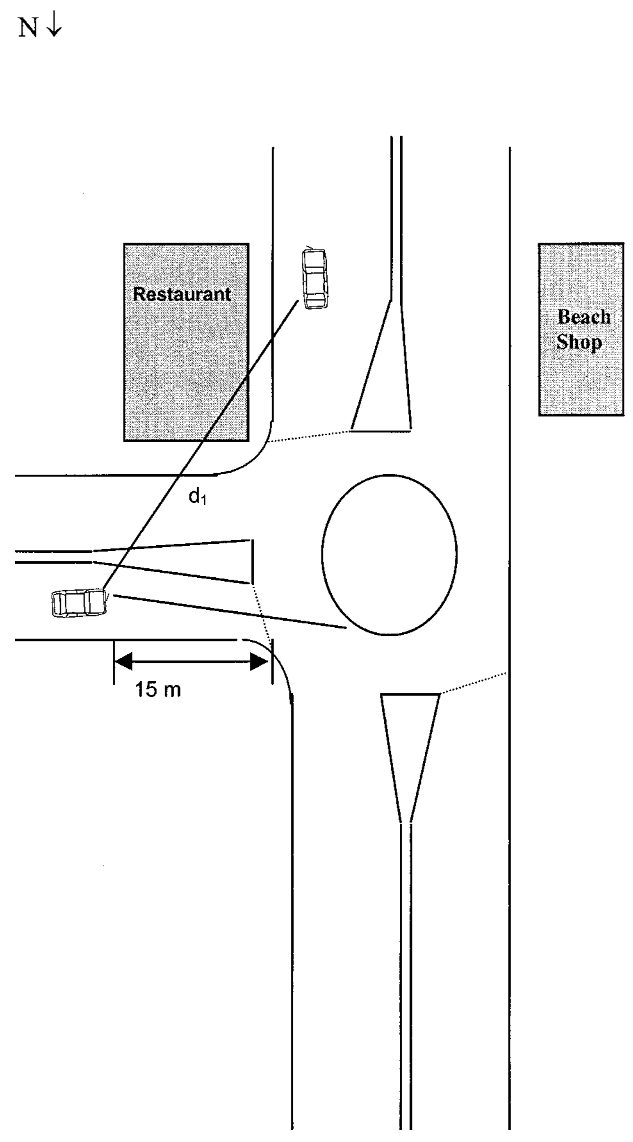

FIGURE 5 Sight distance diagram for Bradenton Beach, Florida.

Summerlin, Nevada, Case Study

One question a designer or planner may ask in the preliminary design phase is "How many vehicles do I design for?" Much guidance available for designers to determine the number of entering and circulating lanes is based on "design hourly volumes," yet little guidance is available as to how far into the future these volumes should be projected. Most guides suggest that roundabouts not be designed for v/c ratios of more than 0.75 to 0.80 because they may begin to operate at capacity within a short time with growth in traffic volumes (*1*, *3*). Further, most guides suggest that when the estimated v/c ratio is near 0.75 to 0.80, an additional lane should be added to account for future traffic volume needs. Little guidance, however, is given as to the lower bound of v/c ratios at which roundabouts should operate. The only guidance identified regarding the design of roundabouts for future traffic needs was found in the British design guide (*8*). This guide suggests using an interim roundabout design, in which the center island essentially covers an extra lane in the circulating roadway, and reducing the center island in size when the extra lane is needed to improve operations (*8*). This design would also apply to roundabout approaches, in which the additional right-of-way required on each approach would not be paved in the interim design but instead paved at a later date when deemed necessary.

While this approach is valid for future development, no guidance is given as to the lower bound of v/c ratios that roundabouts should operate at in the present. This point is not as critical in single-lane roundabout design, in which vehicle paths are restricted through the circulating roadway by the center island and speeds can be controlled by properly designed approach curves. However, in multilane roundabout design, this point is essential to safety performance. Additional lanes in the circulating roadway that operate with little or no traffic allow entering drivers to adjust their travel path to the largest radii through the roundabout, resulting in high speeds through the roundabout. Through site visits, a case study was developed demonstrating the possible safety hazards that can be encountered when v/c ratios are extremely low.

In Summerlin, Nevada, two multilane roundabouts, referred to as the north and south roundabouts, were studied over a 4-day period in 1995. At the time of the observations, the north and south roundabouts carried approximately 850 and 1,025 vehicles per hour during the peak hour, respectively. However, they were designed to carry 3,000 and 6,000 vehicles per hour in the peak hour. This equates to volume to capacity ratios of 0.28 and 0.17, respectively. The following observations were made during the 4-day site visit. These observations demonstrate the problems associated with operating roundabouts with low v/c ratios during the peak and off-peak times.

- Drivers increased their speed on entry.
- Drivers cut across several lanes in the circulating roadway to exit the roundabouts.

• Several drivers entered the roundabouts by making a left turn and proceeding clockwise to exit.

As drivers approached the entries at these two multilane roundabouts, they actually accelerated on the approach instead of decelerating. This was in part because of the lack of traffic within the circulating roadway and in part because of the nonrestrictive travel radius of the roundabouts. Because of low volumes, drivers were not forced to maintain position in their selected lane and instead adjusted their travel path to travel the largest and most comfortable radii. This behavior allowed entering drivers to maintain or increase their approach speeds as they traveled through the roundabouts. Similarly, without many other vehicles in the circulating roadway, drivers often cut across several lanes in the circulating roadway to take their desired exit.

Another driver behavior observed at these two roundabouts was driving the wrong way. Several entering drivers appeared unsure how to make a left turn with the large center island in their path, and they simply chose the shortest route and turned left on entry, despite the use of directional signs on the center island. This behavior was partially influenced by the lack of other drivers to follow and the shear size of the roundabouts (inscribed diameters 61 m and 92 m, respectively), which does not allow drivers to see the entire layout of the intersection.

For these study sites, an interim design that restricted the number of lanes currently operating to one may have helped to reduce the erratic driver behavior observed in the field. Understandably, developers would like to acquire right-of-way sooner rather than later when traffic demands increase; however, too many operational lanes without the traffic demand can affect the operational and safety performance of roundabouts.

FINDINGS AND RECOMMENDATIONS

After reviewing accident reports and making observations at several roundabouts in the United States, the following conclusions were reached:

• Some rural roundabouts experience loss-of-control accidents because of excessive speeds and possible violations of driver expectancy on approach.

• Unbalanced flows can be detrimental to roundabout operations.

• The lack of sight distance can hinder free flow of vehicles into roundabouts.

• Operating roundabouts with extremely low v/c ratios can result in high travel speeds through the roundabout and improper driver behavior.

The findings from this study should help designers, planners, and engineers to understand the basic design requirements for roundabouts and to avoid some of the design errors identified in this paper.

FUTURE RESEARCH NEEDS

While the recently published FHWA informational guide will greatly assist designers, planners, and engineers to understand the principles of design and operation of roundabouts, much of the information in the guide was drawn from international experience with roundabouts. Revisiting the topic of roundabout design will be necessary when sufficient time has passed and more information and data can be drawn from U.S. experiences. In particular, the issues raised in this paper need to be addressed through further research into the minimum v/c at which multilane roundabouts can operate; the maximum inscribed diameter size limits that can be used to avoid confusing unfamiliar drivers during low-traffic-flow periods; and ways to limit the number of loss-of-control accidents at high-speed rural roundabout approaches. These issues, along with many issues relating to human factors and operational performance, need to be addressed through research as more roundabouts are developed in the United States.

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REFERENCES

1. Robinson, B. W., L. Rodegerdts, W. Scarborough, W. Kittelson, R. Troutbeck, W. Brilon, L. Bondzio, K. Courage, M. Kyte, J. Mason, A. Flannery, E. Myers, J. Bunker, and G. Jacquemart. *Roundabouts: An Informational Guide.* Report FHWA-RD-00-067. FHWA, U.S. Department of Transportation, 2000.
2. Zeeger, C. V. *Highway Safety Improvement Program.* Report FHWA-TS-81-218. FHWA, U.S. Department of Transportation, 1981.
3. *Roundabouts—Guide to Traffic Engineering and Practice, Part 6—Roundabouts.* Austroads, Sydney, Australia, 1993.
4. Guichet, B. Roundabouts in France: Development, Safety, Design and Capacity. *Proc., 3rd International Symposium on Intersections Without Traffic Signals,* Portland, Oreg., 1997, pp. 100–105.
5. Giaever, T. Application, Design and Safety of Roundabouts in Norway. The Foundation for Scientific and Industrial Research at the Norwegian Institute of Technology, Norway. Presented in Actes du Seminaire "Giratoires 92," Nates, France, 1992.
6. *Roundabout Design Guidelines.* State of Maryland Department of Transportation, State Highway Administration, 1995.
7. *Draft Guidelines for the Design of Roundabouts.* Transport and Technology Division of the Department of Main Roads, Queensland, Australia, 1997.
8. *Geometric Design of Roundabouts—Design Manual for Roads and Bridges,* Vol. 6: Road Geometry. Report TD 16/93. The Department of Transport, United Kingdom, 1993.

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